Problem 26.3

Find an expression for the accumulated value at time 20 years of payments of \$5 at time 1 year, \$10 at time 2 years, \$15 at time 3 years, and so on, up to \$100 at time 20 years.

Solution

The payments are

$$5, 10, 15, \ldots, 100.$$

Since the payments increase by 5 each year, this is 5 times an increasing annuity-immediate with 20 payments.

The accumulated value at time 20 is

$$5(Is)_{\overline{20}|}.$$

For an increasing annuity-immediate accumulated to time n,

$$(Is)_{\overline{n}|} = \frac{\ddot{s}_{\overline{n}|} - n}{i}.$$

Therefore,

$$(Is)_{\overline{20}|} = \frac{\ddot{s}_{\overline{20}|} - 20}{i}.$$

Thus, the accumulated value is

$$AV = 5\left(\frac{\ddot{s}_{\overline{20}|} - 20}{i}\right).$$

Therefore,

$$\boxed{5(Is)_{\overline{20}|}}$$

or equivalently,

$$\boxed{5\left(\frac{\ddot{s}_{\overline{20}|} - 20}{i}\right)}.$$